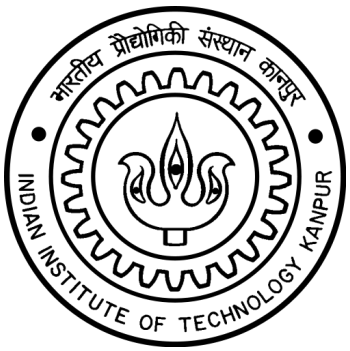


SPA617

RADIO ASTRONOMY LABORATORY

RF Components : RL for Low Pass Filter, Amplifier

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Learning from Viva

What does addition of frequencies in the frequency domain imply in the time domain?

Answer : In a mixer, the two input signals are effectively multiplied in the time domain. For two sinusoidal signals,

$$\cos(2\pi f_{RF}t)\cos(2\pi f_{LO}t) = \frac{1}{2}\left[\cos 2\pi(f_{RF}+f_{LO})t + \cos 2\pi(f_{RF}-f_{LO})t\right].$$

Thus, time-domain multiplication produces the sum and difference frequency components in the frequency domain.

1. Aim

To experimentally characterize the return loss of an RF filter, and to study the performance and nonlinear behaviour of an RF amplifier by measuring its gain, harmonics, 1-dB compression point, and third-order intercept point (TOI).

2. Apparatus

- RF signal generator
- Spectrum analyser
- Digital storage oscilloscope
- RF low-pass filter
- RF amplifier
- DC power supply
- RF Tee connector
- RF power combiner
- 50 Ω load

3. Theory

3.1. What is an amplifier?

An amplifier is an electronic device that increases the amplitude or power of an input signal while preserving its waveform. It uses energy supplied by an external power source to produce a stronger version of the input signal. The ratio of the output signal to the input signal is referred to as the *gain* of the amplifier.

3.2. What is an RF Amplifier?

An RF amplifier is an electronic amplifier designed to increase the power of signals in the radio-frequency range. It is commonly used to strengthen weak RF signals that may come from faint sources.

Unlike a general-purpose amplifier, an RF amplifier is specifically designed to operate efficiently at high frequencies. Its design must account for effects such as impedance matching, bandwidth, and frequency-dependent losses.

Ideally, an amplifier operates linearly, such that the output is simply an amplified version of the input. However, practical RF amplifiers exhibit nonlinear behaviour, particularly at higher input powers, and may be represented as

$$y(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t) + \cdots .$$

here, $x(t)$ are usually cosine functions taking in the incident frequency as the parameter.

These nonlinear terms produce effects such as harmonics of the amplified frequencies.

In the linear operating regime, the gain of the amplifier is given by

$$G_{\text{amp}} = P_{\text{out}} - P_{\text{in}}, \tag{1}$$

where P_{in} and P_{out} are expressed in dBm. In our experiment, the variation of this gain with frequency can be used to characterize the frequency response of the RF amplifier.

3.3. 1-dB Compression Point

The 1-dB compression point is the threshold value of input power, at a fixed frequency, at which the measured gain of the amplifier decreases by 1 dB from its previously constant gain.

This reduction in gain arises due to nonlinear effects that become significant in the high-power regime. As the amplitude of the input signal is increased, the linear relationship between the input and output begins to break down, resulting in gain compression.

Hence, the 1-dB compression point provides an important measure of the maximum input power up to which the amplifier can be considered to operate approximately linearly.

3.4. Third-Order Intercept Point (TOI)

The third-order intercept point (TOI) provides a measure of the extent to which third-order nonlinear effects become significant in an amplifier. When two closely spaced frequencies f_1 and f_2 are simultaneously applied to a nonlinear amplifier, third-order products are generated symmetrically at:

$$f_{\text{IM3,1}} = 2f_1 - f_2 \tag{2}$$

and

$$f_{\text{IM3,2}} = 2f_2 - f_1. \tag{3}$$

As the input power is increased, the output power of the fundamental components increases approximately with a slope of $1:1$, whereas the third-order products increase more rapidly, with a slope of approximately $3:1$.

The fundamental and third-order output powers are plotted against the input power, and their linear trends are extrapolated. The point at which these extrapolated lines intersect is called the *third-order intercept point* (TOI).

A higher TOI indicates that the third-order products remain relatively weak over a larger range of signal powers, corresponding to better amplifier linearity. In practice, the amplifier generally reaches compression before the theoretical TOI is actually reached.

This is of practical importance, since the real signals that are amplified by the RF amplifier contain mixed frequencies. Hence, it is always preferred to know the power that can be amplified, without the non-linear third order terms dominating over the first order terms.

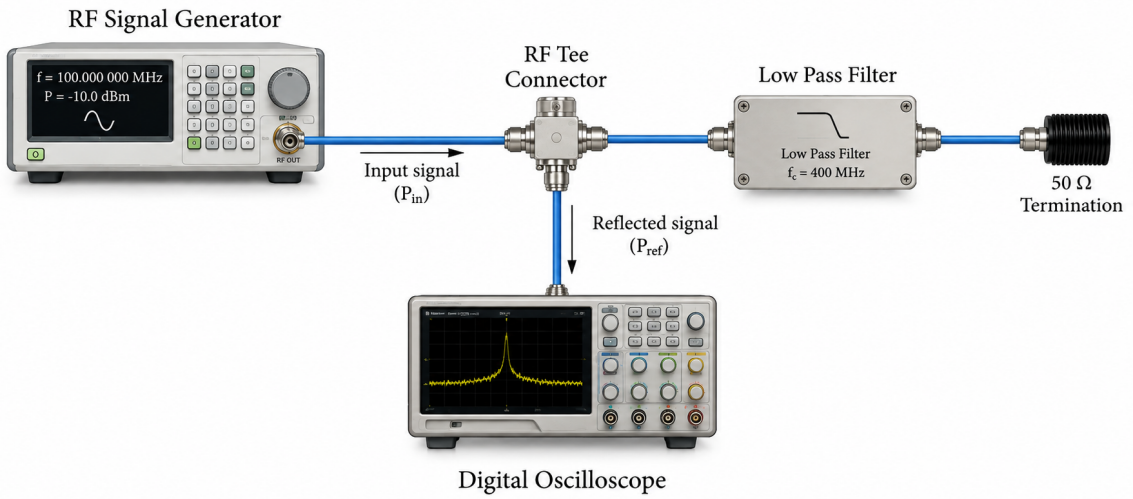


Figure 1: Experimental setup for measuring the reflection coefficient and return loss of the low-pass filter using the standing-wave method.

4. Experimental Setup:

4.1. Calculating the Return loss of the Low-Pass Filter

We use the standard standing-wave formation method to estimate the reflection coefficient and return loss of the low-pass filter. The RF signal is applied through a Tee connector, and the Digital Oscilloscope is used to measure the maximum and minimum voltage amplitudes produced by the interference of the incident and reflected waves. The output of the low-pass filter is terminated with a 50 Ω load to minimize reflections from its output.

4.2. The general setup of Amplifier measurements

For the characterization of the RF amplifier, the amplifier was powered using an external DC power supply connected through the red and black supply wires. The input of the amplifier was connected to the RF signal generator, while its output was connected directly to the spectrum analyser. The input frequency and power were varied as required to study the gain and nonlinear response of the amplifier.

5. Observations and Calculations

5.1. A2 : RL for filter

Table 1: Measured maximum and minimum voltage amplitudes for the filter reflection measurement.

$V_{\max}$ (V)	$V_{\min}$ (V)
1.75	0.987

5.2. Calculation of Reflection Coefficient and Return Loss

The measured maximum and minimum voltage amplitudes were

$$V_{\max} = 1.75 \text{ V}, \qquad V_{\min} = 0.987 \text{ V}.$$

The incident voltage amplitude is

$$V_{\text{inc}} = \frac{V_{\max} + V_{\min}}{2} \tag{4}$$
$$= \frac{1.75 + 0.987}{2} \tag{5}$$
$$= 1.3685 \text{ V}. \tag{6}$$

The reflected voltage amplitude is

$$V_{\text{ref}} = \frac{V_{\max} - V_{\min}}{2} \tag{7}$$
$$= \frac{1.75 - 0.987}{2} \tag{8}$$
$$= 0.3815 \text{ V}. \tag{9}$$

Hence, the magnitude of the reflection coefficient is

$$|\Gamma| = \frac{V_{\text{ref}}}{V_{\text{inc}}} \tag{10}$$
$$= \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \tag{11}$$
$$= 0.279. \tag{12}$$

The return loss is therefore

$$RL = -20 \log_{10} |\Gamma| \tag{13}$$
$$= -20 \log_{10}(0.279) \tag{14}$$
$$\approx 11.10 \text{ dB}. \tag{15}$$

Thus,

$$\boxed{|\Gamma| \approx 0.279} \qquad \text{and} \qquad \boxed{RL \approx 11.10 \text{ dB}}.$$

5.3. Amplifier Measurements

The RF amplifier was characterized by first measuring its gain as a function of frequency at a constant input power. The input frequency was then fixed at 100 MHz and the input power was varied to study gain compression. Finally, a two-tone input was used to observe the third-order intermodulation products and characterize the third-order intercept behaviour of the amplifier.

5.3.1 Gain as a Function of Frequency

The input power was maintained at $P_{\text{in}} = -20$ dBm, while the input frequency was varied from 100 to 600 MHz. The amplifier gain was calculated using

$$G = P_{\text{out}} - P_{\text{in}}. \tag{16}$$

Table 2: Measured amplifier output power and gain as a function of frequency.

Frequency (MHz)	P_{in} (dBm)	P_{out} (dBm)	Gain (dB)
100	-20	2.40	22.40
150	-20	2.07	22.07
200	-20	1.98	21.98
250	-20	1.68	21.68
300	-20	1.70	21.70
350	-20	1.40	21.40
400	-20	1.27	21.27
450	-20	1.21	21.21
500	-20	0.62	20.62
550	-20	0.77	20.77
600	-20	0.46	20.46

5.3.2 Calculation of the 1-dB Compression Point

In the low-power operating region, the amplifier gain remains approximately constant. Taking the average gain over the small-signal region gives

$$G_0 \approx 22.33 \text{ dB}.$$

Hence, the gain corresponding to the 1-dB compression point is

$$G_{1\text{dB}} = G_0 - 1 \approx 21.33 \text{ dB}.$$

From the measured data,

$$G = 21.50 \text{ dB} \quad \text{at} \quad P_{\text{in}} = -6 \text{ dBm},$$

and

$$G = 21.02 \text{ dB} \quad \text{at} \quad P_{\text{in}} = -5 \text{ dBm}.$$

Using linear interpolation,

$$P_{1\text{dB,in}} = -6 + \frac{21.50 - 21.33}{21.50 - 21.02} \tag{17}$$

$$\approx -5.65 \text{ dBm}. \tag{18}$$

Therefore, the input 1-dB compression point is

$$\boxed{P_{1\text{dB,in}} \approx -5.65 \text{ dBm}}.$$

The corresponding output power is

$$P_{1\text{dB,out}} = P_{1\text{dB,in}} + G_{1\text{dB}} \tag{19}$$
$$= -5.65 + 21.33 \tag{20}$$
$$\approx 15.68 \text{ dBm}. \tag{21}$$

Hence,

$$\boxed{P_{1\text{dB,out}} \approx 15.68 \text{ dBm}}.$$

Table 3: Amplifier output power and gain as a function of input power at $f = 100$ MHz.

P_{in} (dBm)	P_{out} (dBm)	Gain (dB)
-20	2.40	22.40
-19	3.30	22.30
-18	4.37	22.37
-17	5.34	22.34
-16	6.25	22.25
-15	7.34	22.34
-14	8.30	22.30
-13	9.20	22.20
-12	10.20	22.20
-11	11.29	22.29
-10	12.20	22.20
-9	13.20	22.20
-8	14.06	22.06
-7	14.85	21.85
-6	15.50	21.50
-5	16.02	21.02
-4	16.40	20.40

5.3.3 Third-Order Intermodulation Measurement

For the third-order intermodulation measurement, two input tones at

$$f_1 = 10 \text{ MHz}, \qquad f_2 = 13 \text{ MHz}$$

were simultaneously applied to the amplifier. The fundamental output power and the power of the adjacent third-order intermodulation product were measured for different values of input power per tone.

5.3.4 Calculation of Third-Order Intercept Point

The linear regions of the measured fundamental and third-order intermodulation (IM3) powers were fitted and extrapolated. The resulting linear fits were

$$P_{\text{fund}} = 0.859 P_{\text{in}} + 17.439 \tag{22}$$

and

Table 4: Measured fundamental and third-order intermodulation powers for the two-tone amplifier test.

P_{in} per tone (dBm)	P_{out} fundamental (dBm)	$P_{3\text{rd}-\text{order}}$ (dBm)
-10	8.75	-27.62
-9	9.66	-24.02
-8	10.78	-20.14
-7	11.44	-17.34
-6	12.25	-14.46
-5	13.08	-11.80
-4	13.83	-9.43
-3	14.54	-7.32
-2	15.10	-5.60
-1	15.70	-4.20
0	16.13	-3.39
1	16.54	-3.01

$$P_{\text{IM3}} = 3.159P_{\text{in}} + 4.466. \tag{23}$$

The measured slopes of approximately 0.86 and 3.16 are reasonably consistent with the theoretically expected slopes of $1:1$ and $3:1$, respectively.

At the third-order intercept point, the extrapolated fundamental and IM3 powers are equal. Therefore,

$$0.859P_{\text{in}} + 17.439 = 3.159P_{\text{in}} + 4.466, \tag{24}$$
$$17.439 - 4.466 = (3.159 - 0.859)P_{\text{in}}, \tag{25}$$
$$P_{\text{in}} = \frac{12.973}{2.300} \tag{26}$$
$$\approx 5.64 \text{ dBm}. \tag{27}$$

Thus, the input third-order intercept point is

$$\boxed{\text{IIP3} \approx 5.64 \text{ dBm}}. \tag{28}$$

Substituting this value into the fundamental linear fit gives

$$P_{\text{out}} = 0.859(5.64) + 17.439 \tag{29}$$
$$\approx 22.29 \text{ dBm}. \tag{30}$$

Hence, the corresponding output third-order intercept point is

$$\boxed{\text{OIP3} \approx 22.29 \text{ dBm}}. \tag{31}$$

Therefore, the experimentally estimated third-order intercept is $(P_{\text{in}}, P_{\text{out}}) \approx (5.64, 22.29)$ dBm.

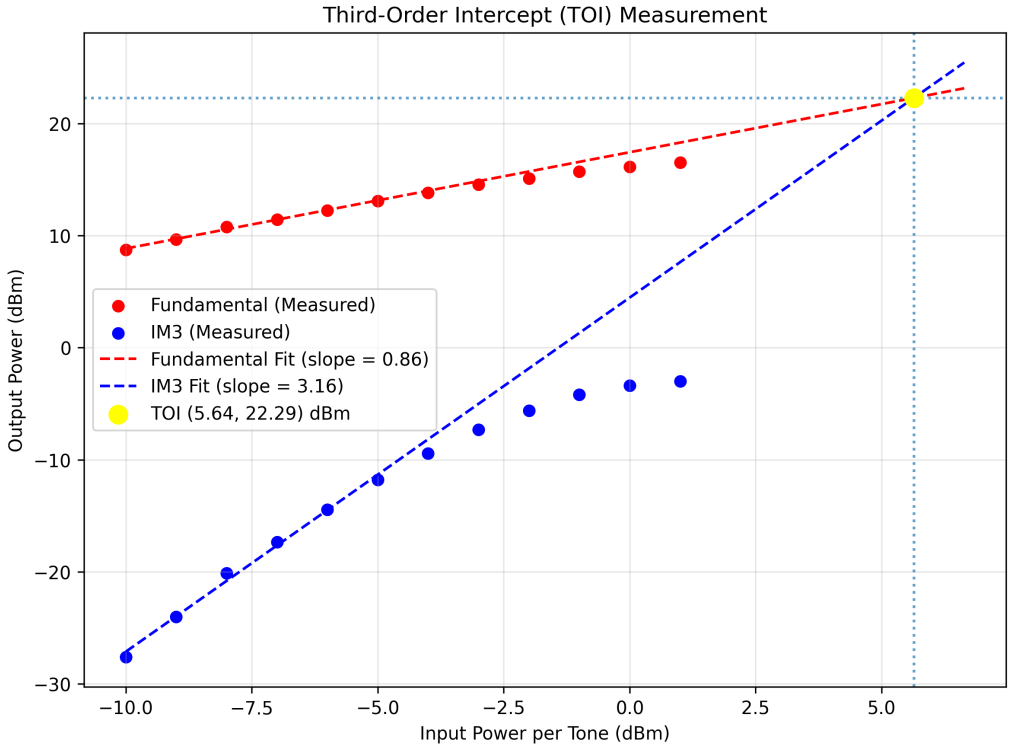


Figure 2: Measured fundamental and third-order intermodulation powers with their linear extrapolations for determination of the third-order intercept point (TOI).

6. Results and Discussion

6.1. Return Loss of the Low-Pass Filter

For the low-pass filter, the measured values $V_{\max} = 1.75$ V and $V_{\min} = 0.987$ V gave a reflection coefficient of

$$|\Gamma| \approx 0.279,$$

corresponding to a return loss of

$$\boxed{RL \approx 11.10 \text{ dB}}.$$

This indicates that the filter is reasonably matched at the measured frequency, although a finite reflected component is still present.

6.2. RF Amplifier

The amplifier exhibited a small-signal gain of approximately 22.3 dB at 100 MHz. As the input power was increased, the gain decreased due to compression. The input 1-dB compression point was estimated to be approximately

$$\boxed{P_{1\text{dB,in}} \approx -5.6 \text{ dBm}}.$$

The two-tone measurement further demonstrated third-order intermodulation. Linear extrapolation of the fundamental and IM3 components gave

$$\boxed{\text{IIP3} \approx 5.64 \text{ dBm}}, \qquad \boxed{\text{OIP3} \approx 22.29 \text{ dBm}}.$$

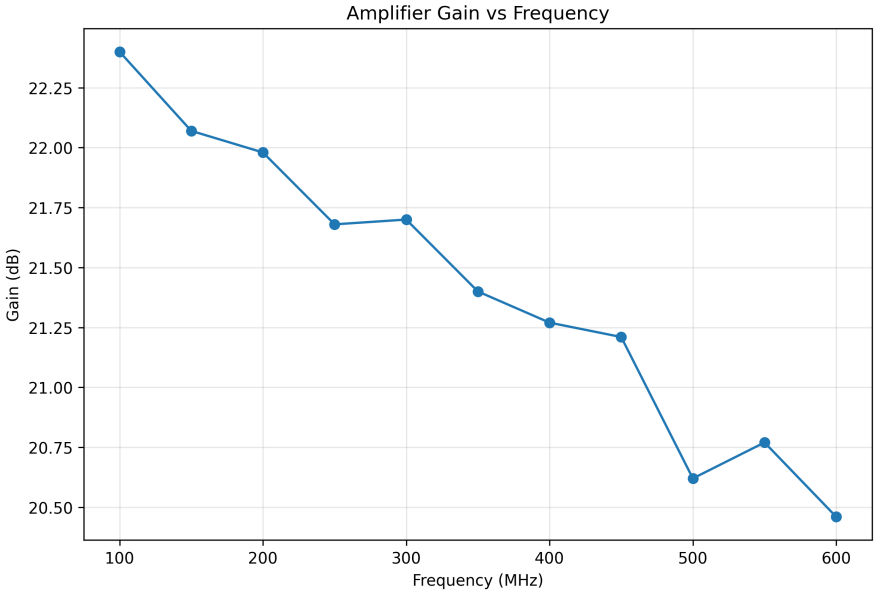


Figure 3: Variation of amplifier gain with input frequency for a constant input power of −20 dBm.

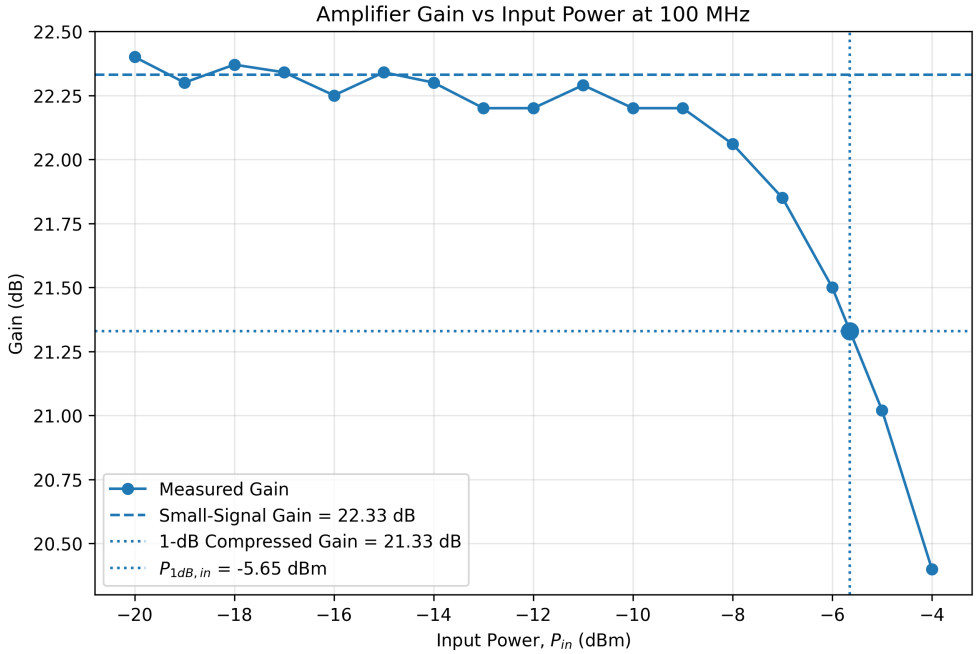


Figure 4: Variation of amplifier gain with input power at $f = 100$ MHz, showing the experimentally determined 1-dB compression point.

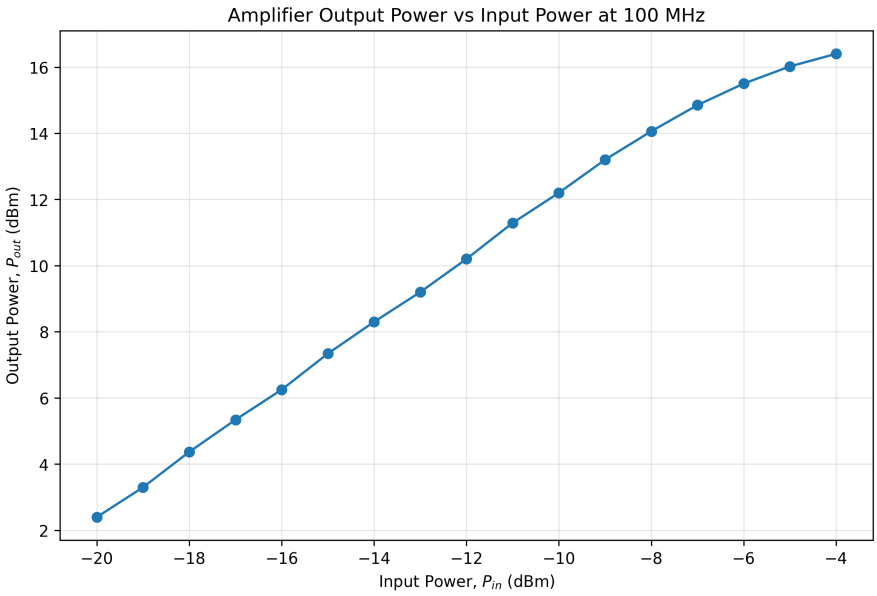


Figure 5: Variation of amplifier output power with input power at a fixed frequency of 100 MHz, showing the onset of gain compression at higher input powers.

AI Declaration

AI tools (primarily ChatGPT) were used as an assistive tool during the preparation of this report. The following uses of AI are acknowledged:

1. **Data representation and LaTeX formatting:** ChatGPT was used to organize the experimentally measured data into clear and readable tables in LaTeX format. All numerical experimental readings used in these tables were obtained during the laboratory experiment.
2. **Plotting and extrapolation:** ChatGPT was used to assist in writing Python code for plotting the experimental data. It was also used to implement the linear fitting and extrapolation of the fundamental and third-order intermodulation components and to determine their intersection for estimation of the third-order intercept point (TOI).
3. **Conceptual understanding:** ChatGPT was used as a learning aid to improve my understanding of the physical significance of the 1-dB compression point and the third-order intercept point (TOI), particularly their relevance in characterizing the nonlinear behaviour of an RF amplifier.
4. **Verification of calculations:** ChatGPT was used to independently verify selected calculations performed using the experimentally measured data, including the reflection coefficient, return loss, amplifier gain, and the calculation of the third-order intercept point.
5. **Experimental setup figure:** AI-based image generation was used to create a schematic representation of the experimental setup used for the low-pass filter reflection and return-loss measurement.
6. **Language refinement:** ChatGPT was used to improve the clarity, grammar, structure, and presentation of the written content throughout the report. The experimental observations, measurements, interpretations, and underlying ideas presented in the report were provided by me.

Declaration by Akshar Kishore